POSTWAR CONFLICTS



Global reach
Fast Combat Support Ship USS *Bridge* leads a flotilla of vessels in support of Operation Enduring Freedom in 2001. Auxiliary vessels such as the *Bridge* enable the US Navy to refuel, rearm, and restock warships wherever they are in the world, extending US naval power across the globe.

IN THE SECOND HALF of the 20th century, the developments in naval tactics and technology evident during World War II continued apace. Big-gun naval vessels were reduced to a shore bombardment role, before virtually disappearing as ever larger aircraft carriers and submarines took center stage. By the 1960s carriers were operating missile-armed jet aircraft from their decks and nuclear-powered, nuclear-armed submarines had become a central part of the Cold War "nuclear deterrent." Surface warships gained a new lease of life through the introduction of shipborne missiles, which not only equipped escort ships for fleet defense but also provided powerful offensive weapons. The days of pitched battles between fleets seemed over, but the relevance of maritime strength to military power projection remained essential, in particular to the global role of the United States, from the Korean War through to modern Middle East conflicts.

NO BIG BATTLES

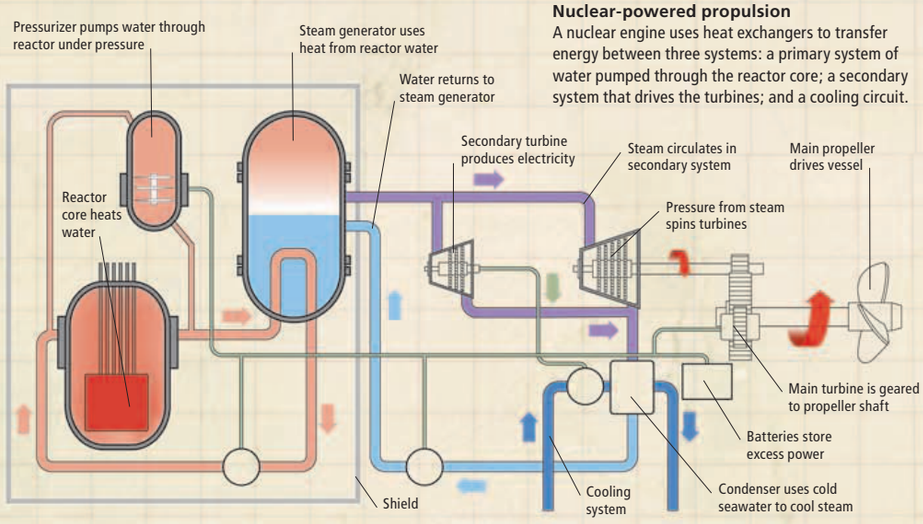
Since the end of World War II there have been no major battles at sea. The end of the war found the United States in possession of what was in practice an unchallenged worldwide naval supremacy. Defeated Germany and Japan were denied any significant naval forces. Britain and France still maintained navies of significant size, but were not potential opponents of the United States. The Soviet Union emerged as a global superpower rival for the United States in the Cold War, but the Soviet navy was not a major force in the first decades after the war. When the Soviet Union did develop a major fleet, by the 1970s, both sides planned for a sea conflict that never happened. Indeed, the Soviets and Americans observed rules to prevent hostile encounters between their navies. Thus the Soviet Union did not intervene with its navy during the Korean War in the 1950s, although the United States and its Allies operated large fleets close to the Soviet Pacific coast, and during the Vietnam War the Americans mostly did not interfere with Soviet seaborne supplies of war material to North Vietnam. With the demise of the Soviet Union in 1991, the United States was left with no credible enemy for a

WEAPONS AND TECHNOLOGY

NUCLEAR POWER

Nuclear propulsion has been in existence for over half a century—the first nuclear-powered vessel, the submarine USS *Nautilus,* was launched in 1955. Nuclear power has some impressive advantages over conventional propulsion systems since nuclear reactors produce a high power output and rarely need refueling. For these reasons, and the fact that nuclear reactors do not require air to operate, nuclear propulsion is ideal for submarines, which spend long periods of time submerged and away from refueling ships or ports.

Nuclear power has also been used in some surface vessels. A basic steam-turbine nuclear reactor works by generating heat through the fission of nuclear material. The heat converts water to steam, and the steam is used to drive a turbine that powers the vessel's propshaft and generators. Nuclear propulsion is not without disadvantage: heavy and expensive structures need to be put in place to contain the radiation, and the reactor core continues to be radioactive for hundreds of years after the ship is decommissioned.



Nuclear-powered propulsion
A nuclear engine uses heat exchangers to transfer energy between three systems: a primary system of water pumped through the reactor core; a secondary system that drives the turbines; and a cooling circuit.

